

GEON CHOI

Ph.D. Student @ KAIST AI

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RESEARCH INTERESTS

My research focuses on **Foundation Models** for **Multimodal Learning**, with an emphasis on **Vision-Language Models (VLMs)** and **Large Language Models (LLMs)** applied to **AI for Healthcare**. I am particularly interested in advancing **Computer Vision** and **Generative Modeling** to build large-scale, real-world systems that bridge cutting-edge foundation model research with high-impact medical applications.

PUBLICATIONS

CHIL 2026	Language: A Benchmark for Structured and Sequential Chest X-ray Interpretation Jong Hak Moon, <u>Geon Choi</u>, Paloma Rabaey, Min Gwan Kim, Hyuk Gi Hong, Jung-Oh Lee, Hangyul Yoon, Eun Woo Doe, Jiyou Kim, Harshita Sharma, Daniel C. Castro, Javier AlvarezValle, Edward Choi <i>Conference on Health, Inference, and Learning, 2026</i> Contribution: Developed an LLM-based framework for structuring radiology reports. Links: [arXiv] [Code]
CVPR 2026	Instruction-Guided Lesion Segmentation for Chest X-rays with Automatically Generated Large-Scale Dataset <u>Geon Choi</u>[*], Hangyul Yoon[*], Hyunju Shin, Hyunki Park, Sang Hoon Seo, Eunho Yang, Edward Choi ([*]: Equal Contribution) <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026</i> Contribution: Co-led the project; designed and implemented the automatic large-scale dataset generation framework. Links: [arXiv] [Code]
NeurIPS 2025 Spotlight	CXReasonBench: A Benchmark for Evaluating Structured Diagnostic Reasoning in Chest X-rays Hyungyung Lee, <u>Geon Choi</u>, Jung-Oh Lee, Hangyul Yoon, Hyuk Gi Hong, Edward Choi <i>Conference on Neural Information Processing Systems (Datasets and Benchmarks Track), 2025</i> Contribution: Built the information-extraction pipeline for X-ray image quality assessment sub-tasks (inclusion, rotation). Links: [arXiv] [Code]
Under Review	CheXpercept: A Benchmark for Evaluating Expert-Level Lesion Perception in Chest X-rays <u>Geon Choi</u>, Hangyul Yoon, Nalee Kim, Jeong Yun Jang, Hyunju Shin, Hyunki Park, Sang Hoon Seo, Edward Choi Contribution: Led the project end-to-end; designed and built the benchmark and conducted all model evaluations.

EDUCATION

2/2026 - Present	Korea Advanced Institute of Science and Technology (KAIST) Ph.D., Artificial Intelligence (Advisor: Prof. Edward Choi)	Seongnam, South Korea
2/2024 - 2/2026	Korea Advanced Institute of Science and Technology (KAIST) M.S., Artificial Intelligence (Advisor: Prof. Edward Choi)	Seongnam, South Korea
2/2020 - 2/2024	Pohang University of Science and Technology (POSTECH) B.S., Electrical Engineering GPA: 4.04/4.3 (Summa Cum Laude) Ranked 2nd in the department of electrical engineering.	Pohang, South Korea
3/2018 - 2/2020	Jeonbuk Science High School Early Graduation Ranked 3rd in class, earning the POSTECH Presidential Fellowship.	Iksan, South Korea

EXPERIENCE

6/2023 - 8/2023	KAIST AI Research Internship (KAIRI) • Biomedicine, Signal Processing and Learning Lab, KAIST (Advisor: Prof. Jongchul Ye) • Research Topic: Generative Models for Solving Inverse Problems in Medical Imaging	Daejeon, South Korea
1/2023 - 6/2023	Undergraduate Research Internship • Bio Optics and Acoustics Lab, POSTECH (Advisor: Prof. Chulhong Kim) • Research Topic: Computer-Aided Diagnosis System for Breast Ultrasonography	Pohang, South Korea

HONORS AND AWARDS

2024 - Present	KAIST Scholarship Tuition Scholarship	KAIST
2023	KEC Scholarship Merit Scholarship for Outstanding EE Student (\$2k)	KEC Foundation
2022 - 2023	National Science & Technology Scholarship Full Tuition Scholarship	KOSAF
2020 - 2021	POSTECH Jigok Scholarship Full Tuition Scholarship	POSTECH
2020 - 2023	POSTECH Presidential Fellowship Merit Fellowship for Exceptional Freshman (\$10k)	POSTECH

TEACHING

2022	Course Assistant for Digital System Design (EECE273) Student Mentor	POSTECH
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SERVICES

2024 - 2025	KoSAIM Summer School Hands-on Session Instructor: taught chest X-ray classification with CNN for medical AI researchers and medical students in South Korea.	KoSAIM
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LANGUAGES

English - professional working proficiency
Korean - native

SKILLS

Programming Languages: Python, C
Deep Learning: PyTorch, HuggingFace Transformers, DeepSpeed, Accelerate
Foundation Models: LLM / VLM training and inference, diffusion models, vLLM
Medical Imaging: pydicom, OpenCV, scikit-image
Tools & Infrastructure: Git, Linux, TensorBoard